

代数学模拟测试 II

期末必考题型总结

■必考题 I: 线性方程组的解结构: 设系数矩阵为 $A = \begin{pmatrix} 1 & a & a & 0 \\ a & 1 & 0 & c \\ a & 0 & 1 & c \\ 0 & c & c & 1 \end{pmatrix}$, $b = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$, 讨论 a, c 取值时的线性方程组解的结构。

答案 由题意, 经过初等变换得到:

$$(\mathbf{A}, \mathbf{b}) = \begin{pmatrix} 1 & a & a & 0 & 1 \\ a & 1 & 0 & c & 0 \\ a & 0 & 1 & c & 0 \\ 0 & c & c & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & a & a & 0 & 1 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 - 2a^2 - 2c^2 & 0 & -a - c \\ 0 & 0 & 2c & 1 & 1 \end{pmatrix} \quad (4.613)$$

所以, 解的情况为:

1. 当 $a^2 + c^2 \neq \frac{1}{2}$ 时, 方程组有唯一解。
2. 当 $a^2 + c^2 = \frac{1}{2}$, $a + c \neq 0$ 时, 方程组无解。
3. 当 $a^2 + c^2 = \frac{1}{2}$, $a + c = 0$ 时, 方程组有无穷多解。解出两组解:

$$(a) \ a = \frac{1}{2}, \ c = -\frac{1}{2} \text{ 方程组的解系为 } k \begin{pmatrix} -1 \\ 1 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 \\ -1 \\ -1 \\ 0 \end{pmatrix}$$

$$(b) a = -\frac{1}{2}, c = \frac{1}{2} \text{ 方程组的解系为 } k \begin{pmatrix} -1 \\ -1 \\ -1 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 \\ 1 \\ 1 \\ 0 \end{pmatrix}$$

■必考题 II: 线性表达: 判断下列各向量组中的向量 β 是否可表示为其余向量的线性组合。若可以, 求其表达式。

$$1. \alpha_1 = (1, 0, 1)^\top, \alpha_2 = (2, 1, 0)^\top, \alpha_3 = (-1, 1, -2)^\top, \beta = (-3, 2, -4)^\top;$$

$$2. \alpha_1 = (1, 3, 2)^\top, \alpha_2 = (3, -1, 1)^\top, \alpha_3 = (-1, 5, 2)^\top, \alpha_4 = (-1, -3, -2)^\top, \beta = (6, 6, 8)^\top.$$

答案

1. 设 $\beta = x_1\alpha_1 + x_2\alpha_2 + x_3\alpha_3$, 即解方程组

$$\begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 1 & 0 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -3 \\ 2 \\ -4 \end{pmatrix}. \quad (4.614)$$

构造增广矩阵并行变换:

$$\begin{aligned} \overline{A} &= \begin{pmatrix} 1 & 2 & -1 & -3 \\ 0 & 1 & 1 & 2 \\ 1 & 0 & -2 & -4 \end{pmatrix} \xrightarrow{r_3-r_1} \begin{pmatrix} 1 & 2 & -1 & -3 \\ 0 & 1 & 1 & 2 \\ 0 & -2 & -1 & -1 \end{pmatrix} \\ &\xrightarrow[r_1-2r_2]{r_3+2r_2} \begin{pmatrix} 1 & 0 & -3 & -7 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 3 \end{pmatrix} \xrightarrow{r_1+3r_3, r_2-r_3} \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{pmatrix}. \end{aligned} \quad (4.615)$$

得唯一解 $x_1 = 2, x_2 = -1, x_3 = 3$, 故

$$\boxed{\beta = 2\alpha_1 - \alpha_2 + 3\alpha_3}. \quad (4.616)$$

2. 设 $\beta = x_1\alpha_1 + x_2\alpha_2 + x_3\alpha_3 + x_4\alpha_4$, 构造增广矩阵:

$$\begin{aligned} \overline{A} &= \begin{pmatrix} 1 & 3 & -1 & -1 & 6 \\ 3 & -1 & 5 & -3 & 6 \\ 2 & 1 & 2 & -2 & 8 \end{pmatrix} \xrightarrow[r_2-3r_1]{r_3-2r_1} \begin{pmatrix} 1 & 3 & -1 & -1 & 6 \\ 0 & -10 & 8 & 0 & -12 \\ 0 & -5 & 4 & 0 & -4 \end{pmatrix} \\ &\xrightarrow{r_3-\frac{1}{2}r_2} \begin{pmatrix} 1 & 3 & -1 & -1 & 6 \\ 0 & -10 & 8 & 0 & -12 \\ 0 & 0 & 0 & 0 & 2 \end{pmatrix}. \end{aligned} \quad (4.617)$$

最后一行对应矛盾方程 $0 = 2$, 故方程组无解。因此 β 不能由 $\alpha_1, \dots, \alpha_4$ 线性表示。

■必考题 III: 线性性质: 若 $\{\alpha_n\}$ 是 n 个线性无关的 n 维向量, 其中, $\alpha_{n+1} = \sum_{i=1}^n k_i \alpha_i$, 其中系数 k_i 全部不为零。证明: $\{\alpha_{n+1}\}$ 向量组中任意 n 个向量都是线性无关的。

答案

- 利用向量组等价证明: 设向量组 (A): $\alpha_1, \alpha_2, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_n, \alpha_{n+1}$, 向量组 (B): $\alpha_1, \alpha_2, \dots, \alpha_{i-1}, \alpha_i, \alpha_{i+1}, \dots, \alpha_n$, 由已知条件可知 (A) 能被 (B) 线性表示。反之, (B) 中向量 $\alpha_1, \alpha_2, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_n$ 显然能由 (A) 线性表示。下证 α_i 能由 (A) 线性表示。由 $\alpha_{n+1} = k_1\alpha_1 + k_2\alpha_2 + \dots + k_i\alpha_i + \dots + k_n\alpha_n$, 而且 k_i 全不为零, 可得

$$\alpha_i = -\frac{1}{k_i} (k_1\alpha_1 + k_2\alpha_2 + \dots + k_{i-1}\alpha_{i-1} + k_{i+1}\alpha_{i+1} + \dots + k_n\alpha_n - \alpha_{n+1})$$

从而 α_i 可由 (A) 线性表示, 所以 (B) 能由 (A) 线性表示。所以向量组 (A) 与向量组 (B) 等价, 而等价的向量。

- 利用定义证明: 设 $\alpha_1, \alpha_2, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_n, \alpha_{n+1}$ 为 $\alpha_1, \alpha_2, \dots, \alpha_n, \alpha_{n+1}$ 中的任意 n 个向量, 其 $1 \leq i \leq n$ 。若有一组数 $l_1, l_2, \dots, l_{i-1}, l_{i+1}$ 使得

$$l_1\alpha_1 + l_2\alpha_2 + \dots + l_{i-1}\alpha_{i-1} + l_{i+1}\alpha_{i+1} + \dots + l_n\alpha_n + l_{n+1}\alpha_{n+1} = 0$$

将 $\alpha_{n+1} = k_1\alpha_1 + k_2\alpha_2 + \dots + k_n\alpha_n$ 代入上式整理可得

$$\begin{aligned} &(l_1 + l_{n+1}k_1)\alpha_1 + (l_2 + l_{n+1}k_2)\alpha_2 + \dots + (l_{i-1} + l_{n+1}k_{i-1})\alpha_{i-1} + l_{n+1}k_i\alpha_i \\ &+ (l_{i+1} + l_{n+1}k_{i+1})\alpha_{i+1} + \dots + (l_n + l_{n+1}k_n)\alpha_n = 0 \end{aligned}$$

因为 $\alpha_1, \alpha_2, \dots, \alpha_n$ 线性无关, 所以有齐次线性方程组

$$\begin{cases} l_1 + l_{n+1}k_1 = 0 \\ l_2 + l_{n+1}k_2 = 0 \\ \dots\dots\dots \\ l_{i-1} + l_{n+1}k_{i-1} = 0 \\ l_{n+1}k_i = 0 \\ l_{i+1} + l_{n+1}k_{i+1} = 0 \\ \dots\dots\dots \\ l_n + l_{n+1}k_n = 0 \end{cases}$$

因为 $l_{n+1}k_i = 0$, 且 $k_i \neq 0$, 所以 $l_{n+1} = 0$, 从而

$$l_1 = l_2 = \dots = l_{i-1} = l_{i+1} = \dots = l_n = 0$$

所以 $\alpha_1, \alpha_2, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_n, \alpha_{n+1}$ 线性无关。若令 $i = n + 1$, 则 $\alpha_1, \alpha_2, \dots, \alpha_n$ 线性无关是已知条件, 所以, $\alpha_1, \alpha_2, \dots, \alpha_n, \alpha_{n+1}$ 中任意 n 个向量都线性无关。

- **利用秩证明:** 设 $\alpha_1, \alpha_2, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_n, \alpha_{n+1}$ 为 $\alpha_1, \alpha_2, \dots, \alpha_n, \alpha_{n+1}$ 中任意 n 个向量, 其中 $\leq i \leq n$, 则

$$(\alpha_1, \alpha_2, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_n, \alpha_{n+1}) \quad (4.618)$$

$$= (\alpha_1, \alpha_2, \dots, \alpha_n) \begin{bmatrix} 1 & 0 & \dots & 0 & 0 & \dots & 0 & k_1 \\ 0 & 1 & \dots & 0 & 0 & \dots & 0 & k_2 \\ \vdots & \vdots & & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 0 & \dots & 0 & k_{i-1} \\ 0 & 0 & \dots & 0 & 0 & \dots & 0 & k_i \\ 0 & 0 & \dots & 0 & 1 & \dots & 0 & k_{i+1} \\ \vdots & \vdots & & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \dots & 0 & 0 & \dots & 1 & k_n \end{bmatrix} \quad (4.619)$$

$$= (\alpha_1, \alpha_2, \dots, \alpha_n) B \quad (4.620)$$

因为 $\alpha_1, \alpha_2, \dots, \alpha_n$ 线性无关,

所以 $r(\alpha_1, \alpha_2, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_n, \alpha_{n+1}) = r(B)$, 而 $|B| = k_1 \neq 0$ 。故 $r(B) = n = r(\alpha_1, \alpha_2, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_n, \alpha_{n+1})$ 。

所以 $\alpha_1, \alpha_2, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_n, \alpha_{n+1}$ 线性无关。证法三组有相同的秩, 并且 $r(B) = n$, 所以 $r(A) = n$, 从而向量组 (A) 中的 n 个向量线性无关。

- **利用行列式证明:** 两个向量组都是 n 维 n 向量的, 所以, 都是方阵, 可以计算行列式, 那么:

$$\det[\alpha_1, \alpha_2, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_{n+1}] \quad (4.621)$$

$$= \det \left[\alpha_1, \alpha_2, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \sum_{i=1}^n k_i \alpha_i \right] \quad (4.622)$$

$$\frac{c_{\alpha_{n+1}} - k_1 c_{\alpha_1} - \dots - k_n c_{\alpha_n}}{\det[\alpha_1, \alpha_2, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, k_i \alpha_i]} \quad (4.623)$$

$$= (-1)^{n-i} \cdot k_i \cdot \det[\alpha_1, \alpha_2, \dots, \alpha_{i-1}, \alpha_i, \dots, \alpha_n] \quad (4.624)$$

显然, 等号两边的向量组具有相同的线性性质, 得证。

■ **必考题 IV: 最大线性无关组:** 求向量组 $\alpha_1 = (1, -2, 0, 3)^\top$, $\alpha_2 = (2, -5, -3, 6)^\top$, $\alpha_3 = (0, 1, 3, 0)^\top$, $\alpha_4 = (2, -1, 4, 7)^\top$, $\alpha_5 = (5, -8, 1, 2)^\top$ 的秩以及一个最大无关组, 并将其余向量表示为这个无关组的线性表示。

答案 将 $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5$ 列排成矩阵进行初等行变换:

$$(\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5) = \begin{bmatrix} 1 & 2 & 0 & 2 & 5 \\ -2 & -5 & 1 & -1 & -8 \\ 0 & -3 & 3 & 4 & 1 \\ 3 & 6 & 0 & -7 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & 2 & 5 \\ 0 & -1 & 1 & 3 & 2 \\ 0 & -3 & 3 & 4 & 1 \\ 0 & 0 & 0 & -13 & -13 \end{bmatrix} \quad (4.625)$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 0 & 2 & 5 \\ 0 & -1 & 1 & 3 & 2 \\ 0 & 0 & 0 & -5 & -5 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & 2 & 5 \\ 0 & -1 & 1 & 3 & 2 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = B. \quad (4.626)$$

因为 B 中有三个非零行, 所以向量组的秩为 3。又因非零行的第一个不等于零的数分别在 1, 2, 4 列, 所以 $\alpha_1, \alpha_2, \alpha_4$ 是极大线性无关组。对矩阵 B 继续作行变换化为最简形式, 即

$$B \rightarrow \begin{bmatrix} 1 & 0 & 2 & 8 & 9 \\ 0 & 1 & -1 & -3 & -2 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (4.627)$$

可得 $\alpha_3 = 2\alpha_1 - \alpha_2, \alpha_5 = \alpha_1 + \alpha_2 + \alpha_4$ 。

■必考题 V: 正交规范化: 利用施密特算法计算如下向量组的正交规范化基底 (标准正交化):

$$1. \alpha_1 = (1, 1, 1)^\top, \alpha_2 = (1, 2, 3)^\top, \alpha_3 = (1, 4, 9)^\top$$

$$2. \alpha_1 = (1, 0, -1, 1)^\top, \alpha_2 = (1, -1, 0, 1)^\top, \alpha_3 = (-1, 1, 1, 0)^\top$$

$$3. \alpha_1 = (1, 1, 1)^\top, \alpha_2 = (0, 1, 1)^\top, \alpha_3 = (0, 0, 1)^\top$$

$$4. \alpha_1 = (1, 2, 1, 0)^\top, \alpha_2 = (1, 1, 3, 1)^\top, \alpha_3 = (1, 0, 5, 2)^\top, \alpha_4 = (2, 1, -2, 3)^\top$$

答案

$$1. \text{ 取 } \beta_1 = \alpha_1 = (1, 1, 1)^\top, \beta_2 = \alpha_2 - \frac{(\alpha_2, \beta_1)}{(\beta_1, \beta_1)}\beta_1 = (-1, 0, 1)^\top,$$

$$\beta_3 = \alpha_3 - \frac{(\alpha_3, \beta_1)}{(\beta_1, \beta_1)}\beta_1 - \frac{(\alpha_3, \beta_2)}{(\beta_2, \beta_2)}\beta_2 = \left(\frac{1}{3}, -\frac{2}{3}, \frac{1}{3}\right)^\top, \quad (4.628)$$

再令

$$\gamma_1 = \frac{\beta_1}{\|\beta_1\|} = \frac{1}{\sqrt{3}}(1, 1, 1)^\top \quad (4.629)$$

$$\gamma_2 = \frac{\beta_2}{\|\beta_2\|} = \frac{1}{\sqrt{2}}(-1, 0, 1)^\top \quad (4.630)$$

$$\gamma_3 = \frac{\beta_3}{\|\beta_3\|} = \frac{1}{\sqrt{6}}(1, -2, 1)^\top \quad (4.631)$$

则 $\gamma_1, \gamma_2, \gamma_3$ 即为所求。

$$2. \text{ 取 } \beta_1 = \alpha_1 = (1, 0, -1, 1)^\top, \beta_2 = \alpha_2 - \frac{(\alpha_2, \beta_1)}{(\beta_1, \beta_1)}\beta_1 = \left(\frac{1}{3}, -1, \frac{2}{3}, \frac{1}{3}\right)^\top,$$

$$\beta_3 = \alpha_3 - \frac{(\alpha_3, \beta_1)}{(\beta_1, \beta_1)}\beta_1 - \frac{(\alpha_3, \beta_2)}{(\beta_2, \beta_2)}\beta_2 = \left(-\frac{1}{5}, \frac{3}{5}, \frac{3}{5}, \frac{4}{5}\right)^\top, \quad (4.632)$$

再令

$$\gamma_1 = \frac{\beta_1}{\|\beta_1\|} = \frac{1}{\sqrt{3}}(1, 0, -1, 1)^\top \quad (4.633)$$

$$\gamma_2 = \frac{\beta_2}{\|\beta_2\|} = \frac{1}{\sqrt{15}}(1, -3, 2, 1)^\top \quad (4.634)$$

$$\gamma_3 = \frac{\beta_3}{\|\beta_3\|} = \frac{1}{\sqrt{35}}(-1, 3, 3, 4)^\top \quad (4.635)$$

则 $\gamma_1, \gamma_2, \gamma_3$ 即为所求。

3.

$$\beta_1 = \alpha_1 = (1, 1, 1)^\top \quad (4.636)$$

$$\beta_2 = \alpha_2 - \frac{(\alpha_2, \beta_1)}{(\beta_1, \beta_1)} \beta_1 = (0, 1, 1)^T - \frac{2}{3}(1, 1, 1)^T = \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)^T, \quad (4.637)$$

$$\beta_3 = \alpha_3 - \frac{(\alpha_3, \beta_1)}{(\beta_1, \beta_1)} \beta_1 - \frac{(\alpha_3, \beta_2)}{(\beta_2, \beta_2)} \beta_2 \quad (4.638)$$

$$= (0, 0, 1)^T - \frac{1}{3}(1, 1, 1)^T - \frac{1}{2}\left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)^T = \left(0, -\frac{1}{2}, \frac{1}{2}\right)^T \quad (4.639)$$

再单位化, 令

$$\varepsilon_1 = \frac{\beta_1}{\|\beta_1\|} = \frac{1}{\sqrt{3}}(1, 1, 1)^T \quad (4.640)$$

$$\varepsilon_2 = \frac{\beta_2}{\|\beta_2\|} = \frac{1}{\sqrt{6}}(-2, 1, 1)^T \quad (4.641)$$

$$\varepsilon_3 = \frac{\beta_3}{\|\beta_3\|} = \frac{1}{\sqrt{2}}(0, -1, 1)^T \quad (4.642)$$

则 $\varepsilon_1, \varepsilon_2, \varepsilon_3$ 即为所求。

4. 由向量组 $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ 构成矩阵 A , 用初等行变换将 A 化为行阶梯形矩阵:

$$(A) = \begin{pmatrix} 1 & 1 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ 1 & 3 & 5 & -2 \\ 0 & 1 & 2 & 3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & -1 & -2 & -3 \\ 0 & 2 & 4 & -4 \\ 0 & 1 & 2 & 3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (4.643)$$

则 $\alpha_1, \alpha_2, \alpha_4$ 是向量组 $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ 的一个极大线性无关组, 所以 $\alpha_1, \alpha_2, \alpha_4$ 是 V 的一组基。将 $\alpha_1, \alpha_2, \alpha_4$ 标准正交化。先正交化, 再单位化, 令

$$\varepsilon_1 = \frac{\beta_1}{\|\beta_1\|} = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix}, \quad \varepsilon_2 = \frac{\beta_2}{\|\beta_2\|} = \frac{1}{\sqrt{6}} \begin{pmatrix} 0 \\ -1 \\ 2 \\ 1 \end{pmatrix}, \quad \varepsilon_3 = \frac{\beta_3}{\|\beta_3\|} = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 0 \\ -1 \\ 2 \end{pmatrix} \quad (4.644)$$

$$\beta_1 = \alpha_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix} \quad (4.645)$$

$$\beta_2 = \alpha_2 - \frac{(\alpha_2, \beta_1)}{(\beta_1, \beta_1)} \beta_1 = \begin{pmatrix} 1 \\ 1 \\ 3 \\ 1 \end{pmatrix} - \frac{6}{6} \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 2 \\ 1 \end{pmatrix}, \quad (4.646)$$

$$\beta_3 = \alpha_4 - \frac{(\alpha_4, \beta_1)}{(\beta_1, \beta_1)} \beta_1 - \frac{(\alpha_4, \beta_2)}{(\beta_2, \beta_2)} \beta_2 = \begin{pmatrix} 2 \\ 1 \\ -2 \\ 3 \end{pmatrix} - \frac{2}{6} \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix} - \frac{-2}{6} \begin{pmatrix} 0 \\ -1 \\ 2 \\ 1 \end{pmatrix} \quad (4.647)$$

$$= \frac{5}{3} \begin{pmatrix} 1 \\ 0 \\ -1 \\ 2 \end{pmatrix}. \quad (4.648)$$

■ **必考题 VI: 矩阵的特征分析:** 求以下矩阵的特征值与特征向量。并问特征向量是否线性无关。

$$1. \begin{pmatrix} 2 & -1 & 2 \\ 5 & -3 & 3 \\ -1 & 0 & -2 \end{pmatrix}$$

$$2. \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}$$

$$3. \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

1.

$$|\mathbf{A} - \lambda \mathbf{E}| = \begin{vmatrix} 2 - \lambda & -1 & 2 \\ 5 & -3 - \lambda & 3 \\ -1 & 0 & -2 - \lambda \end{vmatrix} \quad (4.649)$$

$$\xrightarrow{\underline{\underline{c_3 - (\lambda + 2)c_1}}} \begin{vmatrix} 2 - \lambda & -1 & \lambda^2 - 2 \\ 5 & -3 - \lambda & -7 - 5\lambda \\ -1 & 0 & 0 \end{vmatrix} \quad (4.650)$$

$$\xrightarrow{\underline{\underline{\text{按 } r_3 \text{ 展开}}}} \begin{vmatrix} -1 & \lambda^2 - 2 \\ 3 + \lambda & 7 + 5\lambda \end{vmatrix} \xrightarrow{\frac{c_2 \div (1 + \lambda)}{c_2 - c_1}} (1 + \lambda) \begin{vmatrix} -1 & \lambda - 1 \\ 3 + \lambda & 4 \end{vmatrix} \quad (4.651)$$

$$= -(1 + \lambda)^3, \quad (4.652)$$

所以 $\mathbf{A}$ 的特征值为 $\lambda_1 = \lambda_2 = \lambda_3 = -1$ (三重根). 对于特征值 -1 , 解方程 $(\mathbf{A} + \mathbf{E})x = 0$ 。由

$$\mathbf{A} + \mathbf{E} = \begin{pmatrix} 3 & -1 & 2 \\ 5 & -2 & 3 \\ -1 & 0 & -1 \end{pmatrix} \xrightarrow{r} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

得特征向量 $\mathbf{p} = (-1, -1, 1)^T$; 请读者注意, 在求特征值时, 为方便求解, 一般尽量避免直接用行列式定义。注意, 由于特征向量组之间是线性相关的, 所以, 特征向量构成的矩阵 $\mathbf{U}$ 是不可逆的, 所以, 不能对角化。这道题目说明不是所有的矩阵都可以对角化的, 之后, 我们会研究可对角化的条件。

$$2. |\mathbf{A} - \lambda \mathbf{E}| = \begin{vmatrix} 1 - \lambda & 2 & 3 \\ 2 & 1 - \lambda & 3 \\ 3 & 3 & 6 - \lambda \end{vmatrix} \xrightarrow{\underline{\underline{c_1 - c_2}}} \begin{vmatrix} -(1 + \lambda) & 2 & 3 \\ 1 + \lambda & 1 - \lambda & 3 \\ 0 & 3 & 6 - \lambda \end{vmatrix}$$

$$\xrightarrow{\frac{c_1 \div (1 + \lambda)}{r_2 + r_1}} (1 + \lambda) \begin{vmatrix} -1 & 2 & 3 \\ 0 & 3 - \lambda & 6 \\ 0 & 3 & 6 - \lambda \end{vmatrix} \quad (4.653)$$

$$= -(1 + \lambda) \begin{vmatrix} 3 - \lambda & 6 \\ 3 & 6 - \lambda \end{vmatrix} = -\lambda(\lambda + 1)(\lambda - 9) \quad (4.654)$$

所以 $\mathbf{A}$ 的特征值为 $\lambda_1 = -1, \lambda_2 = 0, \lambda_3 = 9$ 。当 $\lambda_1 = -1$ 时, 解方程 $(\mathbf{A} + \mathbf{E})x = 0$,

由

$$\mathbf{A} + \mathbf{E} = \begin{pmatrix} 2 & 2 & 3 \\ 2 & 2 & 3 \\ 3 & 3 & 7 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 4 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

得对应的特征向量 $\mathbf{p}_1 = (-1, 1, 0)^T$; 当 $\lambda_2 = 0$ 时, 解方程 $\mathbf{A}\mathbf{x} = 0$, 由

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -3 \\ 0 & -3 & -3 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

得对应的特征向量 $\mathbf{p}_2 = (-1, -1, 1)^T$; 当 $\lambda_3 = 9$ 时, 解方程 $(\mathbf{A} - 9\mathbf{E})\mathbf{x} = 0$, 由

$$\begin{aligned} \mathbf{A} - 9\mathbf{E} &= \begin{pmatrix} -8 & 2 & 3 \\ 2 & -8 & 3 \\ 3 & 3 & -3 \end{pmatrix} \sim \begin{pmatrix} -5 & 5 & 0 \\ 5 & -5 & 0 \\ -1 & -1 & 1 \end{pmatrix} \\ &\sim \begin{pmatrix} 0 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & -2 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & -1 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & 0 \end{pmatrix} \end{aligned}$$

得对应的特征向量 $\mathbf{p}_3 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$;

3. 特征多项式为 $|\mathbf{A} - \lambda\mathbf{E}| = (\lambda^2 - 1)^2$, 所以 $\mathbf{A}$ 的特征值为 $\lambda_1 = \lambda_2 = -1, \lambda_3 = \lambda_4 = 1$ 。

当 $\lambda_1 = \lambda_2 = -1$ 时, 解方程 $(\mathbf{A} + \mathbf{E})\mathbf{x} = 0$, 由

$$\mathbf{A} + \mathbf{E} = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

得对应的线性无关特征向量为

$$\mathbf{p}_1 = (0, -1, 1, 0)^T, \quad \mathbf{p}_2 = (-1, 0, 0, 1)^T$$

当 $\lambda_3 = \lambda_4 = 1$ 时, 解方程 $(\mathbf{A} - \mathbf{E})x = 0$, 由

$$\mathbf{A} - \mathbf{E} = \begin{pmatrix} -1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & 0 & -1 \end{pmatrix} \sim \begin{pmatrix} -1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

得对应的线性无关特征向量为

$$\mathbf{p}_3 = (1, 0, 0, 1)^T, \mathbf{p}_4 = (0, 1, 1, 0)^T.$$

■ **必考题 VII: 矩阵对角化:** 求出正交矩阵 $\mathbf{P}$ 使得 $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ 对角化。

$$1. \mathbf{A} = \begin{pmatrix} 2 & -2 & 0 \\ -2 & 1 & -2 \\ 0 & -2 & 0 \end{pmatrix}$$

$$2. \mathbf{A} = \begin{pmatrix} 2 & 2 & -2 \\ 2 & 5 & -4 \\ -2 & -4 & 5 \end{pmatrix}$$

$$3. \mathbf{A} = \begin{pmatrix} 1 & 4 & -2 \\ 0 & -1 & 0 \\ 1 & 2 & -2 \end{pmatrix}$$

$$4. \mathbf{A} = \begin{pmatrix} 1 & -2 & 1 \\ -2 & 1 & 1 \\ 1 & 1 & -2 \end{pmatrix}$$

答案

1. 先求特征值:

$$|\mathbf{A} - \lambda\mathbf{E}| = \begin{vmatrix} 2-\lambda & -2 & 0 \\ -2 & 1-\lambda & -2 \\ 0 & -2 & -\lambda \end{vmatrix} = -\lambda(1-\lambda)(2-\lambda) - 4(2-\lambda)' + 4\lambda \quad (4.655)$$

$$= -\lambda(1-\lambda)(2-\lambda) - 8(1-\lambda) = (1-\lambda)(\lambda-4)(\lambda+2) \quad (4.656)$$

所以 $\mathbf{A}$ 的特征值为 $\lambda_1 = -2, \lambda_2 = 1, \lambda_3 = 4$ 。再求特征向量：对应 $\lambda_1 = -2$, 解方程 $(\mathbf{A} + 2\mathbf{E})x = 0$, 由

$$\mathbf{A} + 2\mathbf{E} = \begin{pmatrix} 4 & -2 & 0 \\ -2 & 3 & -2 \\ 0 & -2 & 2 \end{pmatrix} \xrightarrow{r} \begin{pmatrix} -2 & 3 & -2 \\ 0 & 4 & -4 \\ 0 & 1 & -1 \end{pmatrix} \xrightarrow{r} \begin{pmatrix} -2 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{r} \begin{pmatrix} -2 & 1 & 0 \\ -2 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

得单位特征向量 $\mathbf{p}_1 = \frac{1}{3}(1, 2, 2)^T$ ；对应 $\lambda_2 = 1$, 解方程 $(\mathbf{A} - \mathbf{E})x = 0$, 由

$$\mathbf{A} - \mathbf{E} = \begin{pmatrix} 1 & -2 & 0 \\ -2 & 0 & -2 \\ 0 & -2 & -1 \end{pmatrix} \xrightarrow{r} \begin{pmatrix} 1 & -2 & 0 \\ 0 & -4 & -2 \\ 0 & -2 & -1 \end{pmatrix} \xrightarrow{r} \begin{pmatrix} 1 & -2 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

得单位特征向量 $\mathbf{p}_2 = \frac{1}{3}(2, 1, -2)^T$ ；对应 $\lambda_3 = 4$, 解方程 $(\mathbf{A} - 4\mathbf{E})x = 0$, 由

$$\mathbf{A} - 4\mathbf{E} = \begin{pmatrix} -2 & -2 & 0 \\ -2 & -3 & -2 \\ 0 & -2 & -4 \end{pmatrix} \xrightarrow{r} \begin{pmatrix} 1 & 1 & 0 \\ 0 & -1 & -2 \\ 0 & -2 & -4 \end{pmatrix} \xrightarrow{r} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

得单位特征向量 $\mathbf{p}_3 = \frac{1}{3}(2, -2, 1)^T$ 。

$$\mathbf{P} = \frac{1}{3} \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix},$$

则 $\mathbf{P}$ 是正交矩阵, 且有

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{P}^T\mathbf{A}\mathbf{P} = \begin{pmatrix} -2 & & \\ & 1 & \\ & & 4 \end{pmatrix}.$$

$$\begin{aligned} 2. \det(\mathbf{A} - \lambda\mathbf{E}) &= \begin{vmatrix} 2-\lambda & 2 & -2 \\ 2 & 5-\lambda & -4 \\ -2 & -4 & 5-\lambda \end{vmatrix} \xrightarrow[\substack{r_2+r_3 \\ c_3-c_2}}{\substack{r_2+r_3 \\ c_3-c_2}} \begin{vmatrix} 2-\lambda & 2 & -4 \\ 0 & 1-\lambda & 0 \\ -2 & -4 & 9-\lambda \end{vmatrix} \\ &= (1-\lambda) \begin{vmatrix} 2-\lambda & -4 \\ -2 & 9-\lambda \end{vmatrix} = -(1-\lambda)^2(\lambda-10) \end{aligned}$$

所以 $\mathbf{A}$ 的特征值为 $\lambda_1 = 10, \lambda_2 = \lambda_3 = 1$ (二重根)。对应 $\lambda_1 = 10$, 解方程 $(\mathbf{A} - 10\mathbf{E})x = 0$, 由

$$\mathbf{A} - 10\mathbf{E} = \begin{pmatrix} -8 & 2 & -2 \\ 2 & -5 & -4 \\ -2 & -4 & -5 \end{pmatrix} \xrightarrow{r} \begin{pmatrix} 2 & -5 & -4 \\ 0 & -9 & -9 \\ 0 & -18 & -18 \end{pmatrix} \quad (4.657)$$

$$\sim \begin{pmatrix} 2 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{r} \begin{pmatrix} -2 & 1 & 0 \\ 2 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \quad (4.658)$$

得单位特征向量 $\mathbf{p}_1 = \frac{1}{3}(1, 2, -2)^T$; 对应 $\lambda_2 = \lambda_3 = 1$, 解方程 $(\mathbf{A} - \mathbf{E})x = 0$, 由

$$\mathbf{A} - \mathbf{E} = \begin{pmatrix} 1 & 2 & -2 \\ 2 & 4 & -4 \\ -2 & -4 & 4 \end{pmatrix} \xrightarrow{r} \begin{pmatrix} 1 & 2 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

得线性无关特征向量: $a_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, a_2 = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$. 将 a_1 和 a_2 正交化得

$$b_1 = a_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad b_2 = a_2 - \frac{(b_1, a_2)}{(b_1, b_1)} b_1 = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix}$$

3. $\mathbf{A}$ 的特征多项式为 $|\lambda\mathbf{E} - \mathbf{A}| = \begin{vmatrix} \lambda - 1 & -4 & 2 \\ 0 & \lambda + 1 & 0 \\ -1 & -2 & \lambda + 2 \end{vmatrix} = \lambda(\lambda + 1)^2$, 所以 $\mathbf{A}$ 的特征值为 $\lambda_1 = \lambda_2 = -1, \lambda_3 = 0$.

当 $\lambda_1 = \lambda_2 = -1$ 时, 解 $(\mathbf{A} + \mathbf{E})x = 0$, 得基础解系为 $\mathbf{p}_1 = (-2, 1, 0)^T, \mathbf{p}_2 = (1, 0, 1)^T$.

当 $\lambda_3 = 0$ 时, 解 $\mathbf{A}x = 0$, 得基础解系为 $\mathbf{p}_3 = (2, 0, 1)^T$. 令 $\mathbf{P} = \begin{pmatrix} -2 & 1 & 2 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$, 则

$$P^{-1}AP = \Lambda = \begin{pmatrix} -1 & & \\ & -1 & \\ & & 0 \end{pmatrix}, \text{故 } A = P\Lambda P^{-1}.$$

4.

$$|\lambda E - A| = \begin{vmatrix} \lambda - 1 & 2 & -1 \\ 2 & \lambda - 1 & -1 \\ -1 & -1 & \lambda + 2 \end{vmatrix} = \lambda(\lambda - 3)(\lambda + 3) \quad (4.659)$$

即 $\lambda_1 = 0, \lambda_2 = 3, \lambda_3 = -3$ 解得对应于 $\lambda_1, \lambda_2, \lambda_3$ 的特征向量分别为

$$\alpha_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \alpha_2 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \alpha_3 = \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} \quad (4.660)$$

$$\text{故有 } P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 0 & -2 \end{pmatrix}, \text{使得 } P^{-1}AP = \begin{pmatrix} 0 & & \\ & 3 & \\ & & -3 \end{pmatrix}$$

■必考题 VIII: 方阵幂次: 设 $A = \begin{pmatrix} 1 & 4 & -2 \\ 0 & -1 & 0 \\ 1 & 2 & -2 \end{pmatrix}$, 求 A^{2014} .

答案 先把 A 相似对角化, 因

$$|\lambda E - A| = \begin{vmatrix} \lambda - 1 & -4 & 2 \\ 0 & \lambda + 1 & 0 \\ -1 & -2 & \lambda + 2 \end{vmatrix} = (\lambda + 1)^2 \lambda$$

从而 A 的特征值为: $\lambda_1 = -1$ (二重), $\lambda_2 = 0$. 当 $\lambda_1 = -1$ 时, 求解 $(-E - A)x = 0$, 得基础解系为:

$$\xi_1 = (-2, 1, 0)^T, \quad \xi_2 = (1, 0, 1)^T$$

当 $\lambda_2 = 0$ 时, 求解 $-Ax = 0$, 得基础解系为:

$$\xi_3 = (2, 0, 1)^T$$

令 $P = (\xi_1, \xi_2, \xi_3)$, 则 $P^{-1}AP = \begin{bmatrix} -1 & & \\ & -1 & \\ & & 0 \end{bmatrix}$. 于是 $A = P \begin{bmatrix} -1 & & \\ & -1 & \\ & & 0 \end{bmatrix} P^{-1}$

所以:

$$(A)^{2014} = P \begin{bmatrix} (-1)^{2014} & & \\ & (-1)^{2014} & \\ & & 0^{2014} \end{bmatrix} P^{-1} = P \begin{bmatrix} 1 & & \\ & 1 & \\ & & 0 \end{bmatrix} P^{-1} = -A$$

■ **必考题 IX: 二次型的标准型:** 用矩阵表达如下的二次型, 通过特征分析与配方法, 将如下二次型化为标准形, 并解释 $f(x_1, x_2, x_3) = 1$ 的几何形式。

1. $f(x) = 2x_1^2 + 4x_1x_2 - 4x_1x_3 + 5x_2^2 - 8x_2x_3 + 5x_3^2$

2. $f(x) = x_1^2 + 4x_2^2 + 4x_3^2 - 4x_1x_2 + 4x_1x_3 - 8x_2x_3$

3. $f(x) = -4x_1x_2 + 2x_1x_3 + 2x_2x_3$

4. $f(x) = 2x_1^2 + 3x_2^2 + 3x_3^2 + 4x_2x_3$

5. $f(x) = x_1^2 + x_3^2 + 2x_1x_3 - 2x_2x_3$

6. $f(x) = 3x^2 + 5y^2 + 5z^2 + 4xy - 4xz - 10yz$

7. $f(x) = \sum_{k=1}^m \left(\sum_{p=1}^n a_{kp} x_p \right)^2$

答案

1. **特征分析:** 由题设,

$$f(x_1, x_2, x_3) = 2x_1^2 + 4x_1x_2 - 4x_1x_3 + 5x_2^2 - 8x_2x_3 + 5x_3^2.$$

其二次型的矩阵为:

$$A = \begin{bmatrix} 2 & 2 & -2 \\ 2 & 5 & -4 \\ -2 & -4 & 5 \end{bmatrix}$$

由

$$|\lambda E - A| = \begin{vmatrix} \lambda - 2 & -2 & 2 \\ -2 & \lambda - 5 & 4 \\ 2 & 4 & \lambda - 5 \end{vmatrix} = (\lambda - 1)^2(\lambda - 10),$$

得 $\mathbf{A}$ 的特征值为 1 (二重) 与 10. 对于 $\lambda = 1$, 求解齐次线性方程组 $(\mathbf{E} - \mathbf{A})\mathbf{x} = \mathbf{0}$, 可得该方程组的一个基础解系为

$$\boldsymbol{\alpha}_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \quad \boldsymbol{\alpha}_2 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

所以 $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2$ 是 $\mathbf{A}$ 的属于特征值 1 的两个线性无关的特征向量. 先正交化

$$\boldsymbol{\beta}_1 = \boldsymbol{\alpha}_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \quad \boldsymbol{\beta}_2 = \boldsymbol{\alpha}_2 - \frac{(\boldsymbol{\alpha}_2, \boldsymbol{\beta}_1)}{(\boldsymbol{\beta}_1, \boldsymbol{\beta}_1)} \boldsymbol{\beta}_1 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} - \left(-\frac{4}{5}\right) \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{2}{5} \\ \frac{4}{5} \\ 1 \end{bmatrix}$$

$$\text{再单位化 } \boldsymbol{\eta}_1 = \frac{\boldsymbol{\beta}_1}{\|\boldsymbol{\beta}_1\|} = \begin{bmatrix} \frac{-2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \\ 0 \end{bmatrix}, \quad \boldsymbol{\eta}_2 = \frac{\boldsymbol{\beta}_2}{\|\boldsymbol{\beta}_2\|} = \begin{bmatrix} \frac{2}{3\sqrt{5}} \\ \frac{4}{3\sqrt{5}} \\ \frac{5}{3\sqrt{5}} \end{bmatrix}. \text{ 对于 } \lambda = 10, \text{ 求解齐次线性}$$

方程组 $(10\mathbf{E} - \mathbf{A})\mathbf{x} = \mathbf{0}$, 可求得其基础解系为

$$\boldsymbol{\alpha}_3 = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}, \text{ 从而 } \boldsymbol{\eta}_3 = \frac{\boldsymbol{\alpha}_3}{\|\boldsymbol{\alpha}_3\|} = \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \\ -\frac{2}{3} \end{bmatrix}.$$

$$\text{令 } \mathbf{Q} = (\boldsymbol{\eta}_1, \boldsymbol{\eta}_2, \boldsymbol{\eta}_3) = \begin{bmatrix} \frac{-2}{\sqrt{5}} & \frac{2}{3\sqrt{5}} & \frac{1}{3} \\ \frac{1}{\sqrt{5}} & \frac{4}{3\sqrt{5}} & \frac{2}{3} \\ 0 & \frac{5}{3\sqrt{5}} & -\frac{2}{3} \end{bmatrix}, \quad \text{则有 } \mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 10 \end{bmatrix},$$

即经正交变换 $\mathbf{x} = \mathbf{Q}y$ 得 $f(x_1, x_2, x_3) = y_1^2 + y_2^2 + 10y_3^2$.

配方法

$$f = 2x_1^2 + 5x_2^2 + 5x_3^2 + 4x_1x_2 - 4x_1x_3 - 8x_2x_3 \quad (4.661)$$

$$= 2x_1^2 + 4x_1x_2 - x_3 + 5x_2^2 + 5x_3^2 - 8x_2x_3 \quad (4.662)$$

$$= 2 \left[x_1^2 + 2x_1(x_2 - x_3) + (x_2 - x_3)^2 - (x_2 - x_3)^2 \right] + 5x_2^2 + 5x_3^2 - 8x_2x_3 \quad (4.663)$$

$$= 2 \left[(x_1 + x_2 - x_3)^2 - (x_2 - x_3)^2 \right] + 5x_2^2 + 5x_3^2 - 8x_2x_3 \quad (4.664)$$

$$= 2(x_1 + x_2 - x_3)^2 + 3x_2^2 + 3x_3^2 - 4x_2x_3 \quad (4.665)$$

$$= 2(x_1 + x_2 - x_3)^2 + 3 \left(x_2 - \frac{2}{3}x_3 \right)^2 + \frac{5}{3}x_3^2 \quad (4.666)$$

$$\text{令} \begin{cases} y_1 = x_1 + x_2 - x_3, \\ y_2 = x_2 - \frac{2}{3}x_3, \\ y_3 = x_3, \end{cases} \quad \text{则标准形为} \quad f = 2y_1^2 + 3y_2^2 + \frac{5}{3}y_3^2.$$

2. 由 $\mathbf{A}$ 的矩阵以及特征多项式, 得到其特征值 $\lambda_1 = 0$ (2 重), $\lambda_2 = 9$ (1 重)。对于特征值 $\lambda_1 = 0$, 可得对应的特征向量 $\xi_1 = (2, 1, 0)^\top$, $\xi_2 = (-2, 0, 1)^\top$ 。对于特征值 $\lambda_3 = 9$, 可得对应的特征向量 $\xi_3 = (1/2, -1, 1)^\top$ 。因为三个特征向量是正交关系, 所以, 将 ξ_1, ξ_2, ξ_3 单位化, 得

$$\eta_1 = \left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}, 0 \right)^\top, \quad \eta_2 = \left(-\frac{2}{\sqrt{5}}, 0, \frac{1}{\sqrt{5}} \right)^\top, \quad \eta_3 = \left(\frac{1}{3}, -\frac{2}{3}, \frac{2}{3} \right)^\top$$

取

$$\mathbf{Q} = \begin{bmatrix} \frac{2}{\sqrt{5}} & -\frac{2}{\sqrt{5}} & \frac{1}{3} \\ \frac{1}{\sqrt{5}} & 0 & -\frac{2}{3} \\ 0 & \frac{1}{\sqrt{5}} & \frac{2}{3} \end{bmatrix}$$

则 $\mathbf{Q}$ 是正交矩阵。于是, 二次型 f 可通过正交变换

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \mathbf{Q} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

化为标准形 y_3^2 , 这是两条平行的直线。

3. 二次型的矩阵为: $\mathbf{A} = \begin{bmatrix} 0 & -2 & 1 \\ -2 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. 由

$$\begin{aligned} |\lambda \mathbf{E} - \mathbf{A}| &= \begin{vmatrix} \lambda & 2 & -1 \\ 2 & \lambda & -1 \\ -1 & -1 & \lambda \end{vmatrix} \\ &= (\lambda - 2)(\lambda^2 + 2\lambda - 2) = (\lambda - 2)(\lambda + 1 + \sqrt{3})(\lambda + 1 - \sqrt{3}), \end{aligned}$$

可求得 $\mathbf{A}$ 的特征值为 $2, -1 + \sqrt{3}, -1 - \sqrt{3}$. 当 $\lambda = 2$ 时, 解齐次线性方程组 $(2\mathbf{E} - \mathbf{A})\mathbf{x} =$

$\mathbf{0}$, 可求得其一个基础解系为 $\alpha_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$, 从而 α_1 是 $\mathbf{A}$ 的属于特征值 2 的一个线

性无关的特征向量. 将其单位化得 $\eta_1 = \frac{\alpha_1}{\|\alpha_1\|} = \begin{bmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}$; 当 $\lambda = -1 + \sqrt{3}$ 时, 解齐次线

性方程组 $[(-1 + \sqrt{3})\mathbf{E} - \mathbf{A}]\mathbf{x} = \mathbf{0}$, 可求得其一个基础解系为 $\alpha_2 = \begin{bmatrix} 1 \\ 1 \\ \sqrt{3} + 1 \end{bmatrix}$, 从而

α_2 是 $\mathbf{A}$ 的属于特征值 $-1 + \sqrt{3}$ 的一个特征向量, 将其单位化得

$$\eta_2 = \frac{1}{\|\alpha_2\|} \alpha_2 = \begin{bmatrix} \frac{1}{\sqrt{6+2\sqrt{3}}} \\ \frac{1}{\sqrt{6+2\sqrt{3}}} \\ \frac{\sqrt{3}+1}{\sqrt{6+2\sqrt{3}}} \end{bmatrix}$$

当 $\lambda = -1 - \sqrt{3}$ 时, 解齐次线性方程组 $[(-1 - \sqrt{3})\mathbf{E} - \mathbf{A}]\mathbf{x} = \mathbf{0}$, 可求得其一个基础解

系为 $\alpha_3 = \begin{bmatrix} -1 \\ -1 \\ \sqrt{3} - 1 \end{bmatrix}$, 从而 α_3 是 $\mathbf{A}$ 的属于特征值 $-1 - \sqrt{3}$ 的一个特征向量, 将其单位化得

$$\eta_3 = \frac{1}{\|\alpha_3\|} \alpha_3 = \begin{bmatrix} \frac{-1}{\sqrt{6-2\sqrt{3}}} \\ \frac{-1}{\sqrt{6-2\sqrt{3}}} \\ \frac{\sqrt{3}-1}{\sqrt{6-2\sqrt{3}}} \end{bmatrix}.$$

$$\text{令 } \mathbf{Q} = (\eta_1, \eta_2, \eta_3) = \begin{bmatrix} \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6+2\sqrt{3}}} & \frac{-1}{\sqrt{6-2\sqrt{3}}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6+2\sqrt{3}}} & \frac{-1}{\sqrt{6-2\sqrt{3}}} \\ 0 & \frac{\sqrt{3}+1}{\sqrt{6+2\sqrt{3}}} & \frac{\sqrt{3}-1}{\sqrt{6-2\sqrt{3}}} \end{bmatrix}$$

则有 $\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 + \sqrt{3} & 0 \\ 0 & 0 & -1 - \sqrt{3} \end{bmatrix}$, 即经正交变换 $x = \mathbf{Q}y$ 得

$$f(x_1, x_2, x_3) = 2y_1^2 + (-1 + \sqrt{3})y_2^2 + (-1 - \sqrt{3})y_3^2$$

4. 二次型 f 的矩阵为 $\mathbf{A} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 2 \\ 0 & 2 & 3 \end{pmatrix}$,

$$|\mathbf{A} - \lambda \mathbf{E}| = \begin{vmatrix} 2-\lambda & 0 & 0 \\ 0 & 3-\lambda & 2 \\ 0 & 2 & 3-\lambda \end{vmatrix} = (2-\lambda) \begin{vmatrix} 3-\lambda & 2 \\ 2 & 3-\lambda \end{vmatrix} \\ = (2-\lambda)(\lambda-1)(\lambda-5)$$

$\mathbf{A}$ 的特征值为 $\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 5$. 对应特征值 $\lambda_1 = 1$, 解方程 $(\mathbf{A} - \mathbf{E})x = 0$, 由

$$\mathbf{A} - \mathbf{E} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 2 & 2 \end{pmatrix} \sim_r \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

对应特征值 $\lambda_2 = 2$, 解方程 $(\mathbf{A} - 2\mathbf{E})x = 0$, 由得单位特征向量 $\mathbf{p}_2 = (1, 0, 0)^\top$; 对应特征值 $\lambda_3 = 5$, 解方程 $(\mathbf{A} - 5\mathbf{E})x = 0$, 由

$$\mathbf{A} - 5\mathbf{E} = \begin{pmatrix} -3 & 0 & 0 \\ 0 & -2 & 2 \\ 0 & 2 & -2 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

得单位特征向量 $\mathbf{A} - \mathbf{p}_3 = \frac{1}{\sqrt{2}}(0, 1, 1)^\top$. 令 $\mathbf{P} = (\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3) = \begin{pmatrix} 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix}$, 则

$\mathbf{P}$ 为正交矩阵, 再作正交变换 $\mathbf{x} = \mathbf{P}\mathbf{y}$, 即 $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$ 或

$$\begin{cases} x_1 = y_2, \\ x_2 = -\frac{1}{\sqrt{2}}y_1 + \frac{1}{\sqrt{2}}y_3, \text{ 便把 } f \text{ 化为标准形} \\ x_3 = \frac{1}{\sqrt{2}}y_1 + \frac{1}{\sqrt{2}}y_3, \end{cases}$$

$$f = y_1^2 + 2y_2^2 + 5y_3^2;$$

5. 二次型的矩阵为 $\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 1 \end{pmatrix}$, 它的特征多项式为

$$\begin{aligned} |\mathbf{A} - \lambda \mathbf{E}| &= \begin{vmatrix} 1-\lambda & 1 & 0 \\ 1 & -\lambda & -1 \\ 0 & -1 & 1-\lambda \end{vmatrix} \\ &\stackrel{\substack{r_1+r_3 \\ r_1 \div (1-\lambda)}}{=} (1-\lambda) \begin{vmatrix} 1 & 0 & 1 \\ 1 & -\lambda & -1 \\ 0 & -1 & 1-\lambda \end{vmatrix} \\ &\stackrel{\substack{c_3-c_1 \\ c_3 \div (1-\lambda)}}{=} (1-\lambda) \begin{vmatrix} 1 & 0 & 0 \\ 1 & -\lambda & -2 \\ 0 & -1 & 1-\lambda \end{vmatrix} \\ &= (1-\lambda) \begin{vmatrix} -\lambda & -2 \\ -1 & 1-\lambda \end{vmatrix} = -(\lambda-2)(\lambda-1)(\lambda+1), \end{aligned}$$

所以 $\mathbf{A}$ 的特征值为 $\lambda_1 = 2, \lambda_2 = 1, \lambda_3 = -1$. 对应 $\lambda_1 = 2$, 解方程 $(\mathbf{A} - 2\mathbf{E})x = 0$, 由

$$\mathbf{A} - 2\mathbf{E} = \begin{pmatrix} -1 & 1 & 0 \\ 1 & -2 & -1 \\ 0 & -1 & -1 \end{pmatrix} \stackrel{r}{\sim} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

得单位特征向量 $p_1 = \frac{1}{\sqrt{3}}(1, 1, -1)^T$; 对应 $\lambda_2 = 1$, 解方程 $(\mathbf{A} - \mathbf{E})x = 0$, 由

$$\mathbf{A} - \mathbf{E} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -1 & -1 \\ 0 & -1 & 0 \end{pmatrix} \approx \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

得单位特征向量 $p_2 = \frac{1}{\sqrt{2}}(1, 0, 1)^T$; 对应 $\lambda_3 = -1$, 解方程 $(\mathbf{A} + \mathbf{E})x = 0$, 由

$$\mathbf{A} + \mathbf{E} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & -1 \\ 0 & -1 & 2 \end{pmatrix} \stackrel{r}{\sim} \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 0 & -1 & 2 \end{pmatrix} \stackrel{r}{\sim} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix}$$

得单位特征向量 $\mathbf{p}_3 = \frac{1}{\sqrt{6}}(-1, 2, 1)^T$. 令 $\mathbf{P} = (\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3)$, 则 $\mathbf{P}$ 为正交矩阵, 再作正交变换 $\mathbf{x} = \mathbf{P}\mathbf{y}$, 即

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix},$$

则化 f 为标准形: $f = 2y_1^2 + y_2^2 - y_3^2$.

6. 记二次曲面为 $f = 1$, 则 f 为二次型, 它的矩阵为

$$\mathbf{A} = \begin{pmatrix} 3 & 2 & -2 \\ 2 & 5 & -5 \\ -2 & -5 & 5 \end{pmatrix}$$

由

$$\begin{aligned} |\mathbf{A} - \lambda \mathbf{E}| &= \begin{vmatrix} 3-\lambda & 2 & -2 \\ 2 & 5-\lambda & -5 \\ -2 & -5 & 5-\lambda \end{vmatrix} \xrightarrow[r_3 \div (-\lambda)]{r_3 + r_2} (-\lambda) \begin{vmatrix} 3-\lambda & 2 & -2 \\ 2 & 5-\lambda & -5 \\ 0 & 1 & 1 \end{vmatrix} \\ &\stackrel{c_2 \leftarrow c_3}{=} (-\lambda) \begin{vmatrix} 3-\lambda & 4 & -2 \\ 2 & 10-\lambda & -5 \\ 0 & 0 & 1 \end{vmatrix} = (-\lambda) \begin{vmatrix} 3-\lambda & 4 \\ 2 & 10-\lambda \end{vmatrix} \\ &= (-\lambda)(\lambda-2)(\lambda-11) \end{aligned}$$

得 $\mathbf{A}$ 的特征值为 $\lambda_1 = 0, \lambda_2 = 2, \lambda_3 = 11$. 对应于 $\lambda_1 = 0$, 解方程 $\mathbf{A}\mathbf{x} = 0$, 由

$$\mathbf{A} = \begin{pmatrix} 3 & 2 & -2 \\ 2 & 5 & -5 \\ -2 & -5 & 5 \end{pmatrix} \xrightarrow[r_2 - 2r_1]{r_1 - r_2, r_3 + r_2} \begin{pmatrix} 1 & -3 & 3 \\ 0 & 11 & -11 \\ 0 & 0 & 0 \end{pmatrix} \stackrel{r}{\sim} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

得单位特征向量 $\mathbf{p}_1 = \frac{1}{\sqrt{2}}(0, 1, 1)^T$; 对应于特征值 $\lambda_2 = 2$, 解方程 $(\mathbf{A} - 2\mathbf{E})\mathbf{x} = 0$. 由

$$\mathbf{A} - 2\mathbf{E} = \begin{pmatrix} 1 & 2 & -2 \\ 2 & 3 & -5 \\ -2 & -5 & 3 \end{pmatrix} \stackrel{r}{\sim} \begin{pmatrix} 1 & 2 & -2 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \stackrel{r}{\sim} \begin{pmatrix} 1 & 0 & -4 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

得单位特征向量 $\mathbf{p}_2 = \frac{1}{3\sqrt{2}}(4, -1, 1)^T$; 对应于特征值 $\lambda_3 = 11$, 解方程 $(\mathbf{A} - 11\mathbf{E})\mathbf{x} = 0$. 由

$$\mathbf{A} - 11\mathbf{E} = \begin{pmatrix} -8 & 2 & -2 \\ 2 & -6 & -5 \\ -2 & -5 & -6 \end{pmatrix} \sim \begin{pmatrix} 2 & -6 & -5 \\ 0 & -22 & -22 \\ 0 & -11 & -11 \end{pmatrix} \sim \begin{pmatrix} 2 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{\varepsilon} \begin{pmatrix} 2 & 0 & 1 \\ -2 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

得单位特征向量 $\mathbf{p}_3 = \frac{1}{3}(1, 2, -2)^T$. 令 $\mathbf{P} = (\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3) = \begin{pmatrix} 0 & \frac{4}{3\sqrt{2}} & \frac{1}{3} \\ \frac{1}{\sqrt{2}} & \frac{-1}{3\sqrt{2}} & \frac{2}{3} \\ \frac{1}{\sqrt{2}} & \frac{1}{3\sqrt{2}} & -\frac{2}{3} \end{pmatrix}$, 则 $\mathbf{P}$

为正交矩阵, 并且正交变换

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 & \frac{4}{3\sqrt{2}} & \frac{1}{3} \\ \frac{1}{\sqrt{2}} & \frac{-1}{3\sqrt{2}} & \frac{2}{3} \\ \frac{1}{\sqrt{2}} & \frac{1}{3\sqrt{2}} & -\frac{2}{3} \end{pmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix} \quad \text{或} \quad \begin{cases} x = \frac{4}{3\sqrt{2}}v + \frac{1}{3}w, \\ y = \frac{1}{\sqrt{2}}u - \frac{1}{3\sqrt{2}}v + \frac{2}{3}w, \\ z = \frac{1}{\sqrt{2}}u + \frac{1}{3\sqrt{2}}v - \frac{2}{3}w \end{cases}$$

所以, 标准方程为 $2v^2 + 11w^2 = 1$.

7. 二次型为: $f(x) = \sum_{k=1}^m \left(\sum_{p=1}^n a_{kp} x_p \right)^2$, 那么, 我们得到规范型 $y_k = \sum_{p=1}^n a_{kp} x_p$, 则坐标

变换为: $f(x) = \sum_{k=1}^m y_k^2$. 可见, 这是一个椭球. 要求活用二次型分析的特性, 而不是遵循算法, 来求解这道题目。